

Response Letter

Submission PDR-10-2025-3881 for Population and Development Review Response to Reviewers

Dear Dr. Raya Muttarak and Dr. Joshua Wilde,

Thank you for taking the time to evaluate our manuscript, “Structured Inequality, Uncertain Lifespans: Demographic Perspectives on Predicting Individual-level Longevity.” We are grateful to the reviewers and the editorial team for their careful read of the manuscript and suggestions for improvement. We have revised the manuscript in line with all comments of the reviewers.

Below, we give a point-by-point response (in blue) to each suggestion.

Response to comments from Reviewer 1

The main contribution of this paper is to show the low predictive power of sociodemographic characteristics on longevity. As discussed in the paper, this finding aligns with existing studies of the proportion of the variance explained by single SES variables based on cross-sectional life table analysis. Nevertheless, beyond confirming this previously known low explained variance in longevity, the paper is important because it alleviates two major concerns in the literature about why this might be: (1) it is not because of synthetic cohort assumptions or within-cohort mortality selection – this study shows that it also holds up for prospective monitoring of real cohorts; and (2) is not because a single SES variable is not capturing enough variation—even with multiple covariates mortality prediction is weak, simply because within-group variation in longevity is enormous. In other words, this paper confirms that mortality is highly stochastic.

I don’t have any substantive concerns. The paper is beautifully written, and the frontend is especially thoughtful, integrating multiple strands of the literature. In fact, it was so well done that the back end of the paper felt a little disappointing by comparison (more on that later). The analysis is sound, and even if the data has some limitations (e.g. not capturing the early life course, limited variables, likely some linkage failures, generalizability across cohorts/contexts), the advantages of the cohort design likely outweigh the drawbacks. I don’t see that any of these limitations would invalidate the main conclusions. With perfect detailed data covering the full lifespan of these cohorts it might be possible to triple or quadruple the predictive power of the covariates. Even still, predictability would be low, which I see as the main point.

[We thank the reviewer for their careful reading of our manuscript and for sharing these excellent comments.](#)

I only have the following minor suggestions:

1. It could be stressed more that this paper is assessing the predictive power of a set of covariates among 30-year-olds, where the variation across individuals in their set of covariates might not be as meaningful for mortality as it would be at other life course stages. I'm not sure, but maybe there's a literature on this. It might also be that it's not so much the set of fixed covariates themselves that are predictive of mortality, but rather changes in certain of these covariates over the life course, which cannot be tested with this data but could be discussed. For example, if we could pick these individuals up at a later census, it's conceivable that changes in marital status, income, rurality, etc. could substantially improve mortality prediction. It's notable that marital status doesn't even show up among the 15 top predictors, possibly because marriage rates were high for this cohort and marital dissolution or early widowhood at a later age might be the more important mortality signal.

Thank you for the suggestion to emphasize this point. We have expanded our discussion of this:

“Several limitations and opportunities for future research warrant discussion. Our main analysis is restricted to men born in 1910, observed in the 1940 Census, and dying between ages 65 and 95. While this design enables a cohort-based approach to lifespan prediction, it excludes early deaths before age 65; nevertheless, approximately 65% of this cohort died within the observed window (Breen and Osborne 2022). Our predictors are measured cross-sectionally in early adulthood at age 29 or 30, and certain predictors such as income, occupation, or marital status may change over the life course. As a result, we cannot assess how life course trajectories or covariates measured at older ages might alter predictability, even though such measures could plausibly carry stronger mortality signals (e.g., marital dissolution in later life). Longitudinal life-course data may improve predictive performance relative to our baseline; however, even substantial improvement—for example, a doubling or tripling of explanatory power—would still imply relatively low overall predictability. Moreover, several key predictors, including educational attainment, are largely fixed by early adulthood for this cohort.” **(Page 21)**

We also now state this explicitly in the data section:

“These predictors are measured cross-sectionally at ages 29 or 30 and do not capture subsequent changes over the life course.” **(Page 12)**

2. Keeping with the life course theme, I found Figure S6 to be interesting and surprising and worth further discussion, maybe even inclusion into the main text. I would have expected greater predictive power from observing cohorts at a later stage in their life course. This analysis finds the opposite, and I am wondering why this could be. Does it relate to the set of ages that are later observed? Other reasons?

We agree that this is an interesting empirical question. However, as all cohorts in our data share the same mortality observation window of 1975–2005, each cohort will have deaths observed at a different set of ages (as you suspect). Given this, we decided not to move the figure into the main text, as the substantive takeaways remain ambiguous. We retain the figure in the supplement to illustrate that predictability remains low across all cohorts considered, and we have clarified this limitation in the revised caption:

“The estimated R^2 values from a replication of the main CenSoc-DMF prediction exercise,

repeated separately for each birth cohort from 1900 to 1920. Performance is based on a gradient boosted model (GBM), our top-performing discrete machine-learning algorithm. We caution against overinterpreting trends, as each cohort will have: (1) covariates measured at a different point of the life course; (2) mortality observed for a different range of ages; and (3) cohort-specific exposures across the life course. In particular, directly testing whether covariates observed at earlier or later stages of the life course are more predictive is challenging because differences in predictive performance are also influenced by the different ages for which mortality is observed (e.g., the 1900 cohort has mortality observed from ages 75 to 105, while the cohort of 1920 has mortality observed from ages 55 to 85). Differences in predictive performance among cohorts likely reflect differences in age exposure rather than the intrinsic informativeness of later-life covariates. We include this figure to emphasize that predictability is low across all cohorts considered.” **(Page 41)**

3. On the one hand, the predictive power of one’s set of covariates on longevity is low. On the other hand, between-group inequalities in mortality are not trivial. One finding that is coming out of the mortality prediction literature is that covariates might have ok discriminatory power, to for example predict which of two individuals is likely to live longer (e.g. Bardolatto et al, which you cite). Drawing on this literature, and its implications, would enrich the discussion, particularly around what importance we see more generally for the social determinants of mortality.

Thank you for pointing this out. We agree and have added the following text expanding upon this:

“On the one hand, the predictive power of observed covariates for longevity is limited; on the other, between-group inequalities in mortality remain substantial. Large differences in life expectancy across sociodemographic groups—by race, education, income, or place—can coexist with considerable within-group variation. Such disparities reflect structural inequality, policy-relevant gradients, and historically rooted disadvantages that remain central to mortality demography. Low predictive power should therefore not be equated with low structural inequality; rather, it reinforces mortality demography’s focus on group-level outcomes and highlights the value of frameworks that integrate within- and between-group components to better understand lifespan inequality (Permanyer 2023, Shi 2022).

Limited predictability clarifies how sociodemographic factors operate at the individual level. Sociodemographic characteristics may explain only a small share of individual-level lifespan variance while still generating meaningful differences in mortality risk and enabling discrimination between individuals’ relative survival prospects, especially when age is considered (Badolatto 2025). From this perspective, predictive analyses complement rather than replace traditional approaches by highlighting how structural gradients and individual-level uncertainty coexist. Prediction provides a quantitative lens for assessing the scale at which social determinants determine lifespan without implying deterministic life-course trajectories. Simply put, individual-level lifespan predictability is low because the between-group inequality is dominated by within-group variation (van Raalte et al., 2012; Vaupel, 1988; Caswell, 2023). This does not diminish the crucial importance of studying between-group disparities, which remain central and policy-relevant.” **(Page 19-20)**

4. The ties to uncertainty demography is interesting and could be further developed. It's worth looking at the recent PDR paper by Sasson (2025), who makes a lot of these same arguments. The paper by Aburto et al. (2023) argues that the implications of lifetime uncertainty might be particularly salient in violent contexts.

We thank the reviewer for this valuable suggestion to further develop connections with uncertainty demography. We agree and have added the following text to our discussion:

“These findings can also be situated within the emerging field of uncertainty demography, which treats uncertainty as an inherent component of population processes deserving of study in its own right (Trinitapoli 2023). Our analysis extends this perspective to mortality by interpreting the predictability of individual lifespans as a measurable form of population-level uncertainty. Low predictability reflects the stochasticity inherent in mortality (uncertainty as an outcome), while that stochasticity is shaped by macro-level forces such as economic volatility, environmental shocks, epidemics, and what might be called life luck—forces that generate uncertainty as a cause of mortality variation. In this sense, uncertainty is not merely noise to be explained away by better covariates and models, but a defining demographic feature that structures both individual and collective experiences of life and death. More broadly, this stochasticity is a key feature not only of lifespan variation but also outcomes such as morbidity and lifetime reproductive output, where variance itself is a substantive object of analysis (Caswell 2023; van Daalen et al. 2022).

Building on this uncertainty demography perspective, our findings highlight how limited predictive performance can coexist with persistent structural inequalities. Rather than viewing limited predictability as a limitation, we interpret it as empirical evidence for a nature–nurture–chance framework, which calls for greater attention to the role of chance in shaping individual lifespans (Sasson 2025). This is operationalized here through high predictive uncertainty in cohort lifespan outcomes: prediction offers a complementary lens for assessing the balance between systematic social determinants and residual uncertainty. This aligns with evidence from studies of genetically identical twins, which find substantial variability in longevity persists even when genetic and environmental factors are closely aligned (Finch and Kirkwood 2000).

This lifespan uncertainty has meaningful implications for lived experiences. Expectations about survival influence human capital investments (Sasson 2016), investment strategies (Abel, 1985; Barro and Friedman, 1977), and health and fertility decisions (Picone, Sloan and Taylor, 2004; Nettle, 2010). Although our study does not directly examine behavioral responses to uncertainty or perception thereof, the stratified patterns of predictability we document suggest that lifetime uncertainty is unevenly distributed across populations. Further, lifespan uncertainty may be especially pronounced and salient in violent or high-risk contexts, where mortality risks both shorten lives and increase unpredictability in the timing of death, complicating long-term planning and decision-making (Aburto et al. 2023).” **(Page 20-21)**

5. On p.18 there are comparisons drawn to the predictability of other life events, from totally different datasets and study designs. While this is acknowledged – to me the designs are too incomparable for this statement: “lifespan is substantially more difficult to predict than material hardship at age 15 ($R^2 = 0.23$) and is most similar to predicting primary caregiver layoff ($R^2 = 0.03$)”. The supplementary analysis itself showed that adding in age and looking at mortality prediction over a 5-year window substantially improved prediction.

Thank you for pointing this out. We agree, and have revised the text accordingly to avoid making direct comparisons:

“Our results also add to a growing literature on the limits of prediction in other social domains (Salganik 2020; Arpino 2022; Dressel 2018; and Zheng 2025). Comparisons across prediction exercises are inevitably limited by differences in data, features, and modeling approaches. Nonetheless, we situate our findings against those from other studies as illustrative points of reference. Rather than offering strict performance benchmarks, these results provide context for the scale of predictability we observed. For instance, Zheng (2025) predicts midlife socioeconomic status using 4,000 covariates from the stratification literature, finding a relatively high R^2 of 0.5. Compared with the life outcome prediction results in Salganik (2020), our reported R^2 is lower than material hardship at age 15 ($R^2 = 0.23$) and most similar to primary caregiver layoff ($R^2 = 0.03$). These comparisons reinforce that lifespan is a unique biosocial outcome shaped by both social and biological processes, with substantial stochasticity that limits predictive accuracy.” **(Page 18)**

Smaller things:

1. “Second, limited predictability reveals how inequality and uncertainty coexist: structural disparities shape the overall distribution of lifespans but remain limited in their ability to determine individual outcomes.” – To me, this sentence goes against the idea that we can only explain a small portion of the variance. I would actually say that the distribution is defined by stochasticity but structural disparities shift and change the skewness.

We agree this sentence is confusing as written, and have rewritten it in line with your helpful suggestion:

“Second, limited predictability reveals how inequality and uncertainty coexist: stochasticity shapes the overall distribution of lifespans, while structural disparities shift its mean and skewness yet remain limited in their ability to determine individual outcomes.” **(Page 3)**

2. First sentence of the background: “inequality in mortality has also increased over time” – This is probably true for the USA, but several European countries with long time series of data have experienced increasing and declining inequalities over time (Zazueta-Borboa et al., 2023), while other countries might not even be experiencing classic mortality gradients (Sudharsanan et al., 2020). It would be good to either rephrase this sentence or make it clear that you are referring to the USA in this section.

We have clarified that this comment is referring to the United States:

“While overall longevity has increased over the course of the 20th century (Dowd, Polizzi and Tilstra, 2025), inequality in mortality has also increased over time in the United States (Preston and Elo, 1995).” **(Page 4)**

3. “Genetic, behavioral, or biomedical predictors fall outside the conceptual scope of our study, which is concerned with the social structuring of mortality.” This is strongly worded, and only holds so long as these factors do not themselves predict the social statuses examined here.

We have revised this sentence in line with your suggestions:

“By design, we focused on sociodemographic variables that capture systemic inequality. Biological and behavioral predictors are not the primary focus of this study, which centers on the social structuring of mortality. Such factors do, however, overlap with the sociodemographic characteristics considered here, and future work could explicitly consider these predictors.” **(Page 22)**

4. What is the number of mothers/fathers/couples in Fig S1?

We have now explicitly referenced this in the caption:

“The CenSoc-Numident cohort of 1910 includes a total of 75,238 individuals (20,167 women and 55,071 men).” **(Page 35)**

Finally, I think it’s wonderful that the authors chose to make the analysis replicable in OSF.

We thank the reviewer for their thoughtful and constructive feedback.

Literature cited:

Aburto, J. M., Di Lego, V., Riffe, T., Kashyap, R., Van Raalte, A., & Torrisi, O. (2023). A global assessment of the impact of violence on lifetime uncertainty. *Science advances*, 9(5), eadd9038.

Sasson, I. (2025). A New Research Agenda for Social Inequalities in Mortality: Challenges and Open Questions. *Population and Development Review*, 51(1), 323-360.

Sudharsanan, N., Zhang, Y., Payne, C. F., Dow, W., & Crimmins, E. (2020). Education and adult mortality in middle-income countries: Surprising gradients in six nationally-representative longitudinal surveys. *SSM-population health*, 12, 100649.

Zazueta-Borboa, J. D., Martikainen, P., Aburto, J. M., Costa, G., Peltonen, R., Zengarini, N., Sizer, A., Kunst, A. E., & Janssen, F. (2023). Reversals in past long-term trends in educational inequalities in life expectancy for selected European countries. *J Epidemiol Community Health*, 77(7), 421-429.

Response to comments from Reviewer 2

This paper is an important contribution to the analysis of lifespan variation. The authors use linked individual records to associate individual lifespans with a large set of social variables, use machine learning to find a model relating lifespan to those variables, and report the fraction of variance (small) explained. The paper is extremely clearly written. It is a valuable contribution not only for the results of this particular study, but for the useful summary of previous work (thank you for Table 1!) on this topic.

I have a few thoughts to share. These are not criticisms, but may be helpful to the authors. As is my practice, I always sign my reviews. Hal Caswell.

We thank the reviewer for their detailed and expert feedback.

1. p. 2, line 19-20. I like this clear statement.

Thank you for the kind words.

2. p. 3, line 36 and also p. 5, line 79. You say that decompositions are limited to a single categorical variate. This is no longer true. Caswell and van Daalen (2025, Vienna Yearbook of Population Research) show how to incorporate multiple factors and their interactions, including a couple of toy examples. A more complete analysis of a four-factor study will be presented at PAA 2026, and will be written up soon.

Thank you for letting us know about this exciting development. We have updated the text accordingly:

“This approach generalizes traditional decompositions of lifespan variation, which in most empirical applications have examined one categorical covariate at a time, by translating multiple covariates into measures of explanatory power at the individual level.” **(Page 3)**

“Most applications of variance decomposition in demography have focused on single categorical factors considered separately; however, recent methodological extensions allow multiple covariates and their interactions to be incorporated within decomposition frameworks (Caswell 2025). Our approach complements this line of work by shifting from variance partitioning toward individual-level prediction using high-dimensional covariate sets.” **(Page 5)**

3. p. 4, line 47-49. Here, or perhaps in the discussion, you may want to mention stochasticity in other demographic outcomes in addition to longevity, including healthy longevity and lifetime reproductive output (LRO). Examples of these are shown in Caswell (2023), which you cite. An example of partition of variance in LRO is given in van Daalen et al. (2022, American Naturalist) for a laboratory experiment on aging in a rotifer. My point is just to emphasize that the argument of Trinitapoli that variance is an important component of analysis, not just error, applies beyond analysis of lifespan.

Thank you for this helpful point. We have now added this additional information:

“These findings can also be situated within the emerging field of uncertainty demography, which treats uncertainty as an inherent component of population processes deserving of study in its own right (Trinitapoli 2023). Our analysis extends this perspective to mortality by interpreting the predictability of individual lifespans as a measurable form of population-level uncertainty. Low predictability reflects the stochasticity inherent in mortality (uncertainty as an outcome), while that stochasticity is shaped by macro-level forces such as economic volatility, environmental shocks, epidemics, and what might be called life luck—forces that generate uncertainty as a cause of mortality variation. In this sense, uncertainty is not merely noise to be explained away by better covariates and models, but a defining demographic feature that structures both individual and collective experiences of life and death. More broadly, this stochasticity is a key feature not only of lifespan variation but also outcomes such as morbidity and lifetime reproductive output, where variance itself is a substantive object of analysis (Caswell 2023; van Daalen et al. 2022).” **(Page 20)**

4. p. 5, footnote 2. This is an important point. There is a difference between variance obtained from actual observations of individual lifetimes (which you do) and variance calculated from the rates estimated for groups (which most decompositions do). [We need a good terminology to discuss these.] In the former case, the variance due to stochasticity may indeed be due to unmeasured heterogeneity. In the latter case, it cannot, because the calculation of variance, from a life table or other Markov chain, explicitly treats every individual within a group as identical. (I discuss this a little on the top of p. 328 of Caswell 2023). I have always said that data on individual lives are (sometimes) available to population biologists, but not to demographers, because we can't put leg bands on people when they are born like we do with albatrosses. But this study, and studies using register data, are changing that.

Thank you for pointing out this important distinction. We have updated the footnote to add this additional information:

“This stochastic variation may reflect truly random processes or latent individual-level determinants. What appears as “luck” in lifespan may instead reflect fine-grained variation in exposures, behaviors, genetics, or life events that are unmeasured. This differs from variance calculated from the rates estimated for groups using life-table or Markov formulations, which assume homogeneity within groups by construction (Caswell 2023).” **(Page 5)**

5. p. 8, line 128. A set of cohort results for variance due to frailty can be found in Hartemink et al. (2017, Theoretical Population Biology 114:107-116), for France, Sweden and Italy.

Thank you for letting us know. We have adapted our original statement accordingly:

“To our knowledge, no prior study has applied a cohort-based prediction framework to quantify lifespan predictability in the United States.” **(Page 8)**

We have also added a footnote further clarifying:

“Prior work has examined cohort patterns in lifespan variance and frailty in several European countries for historical cohorts, finding only a small amount of lifespan variance is due to frailty (Hartemink 2017).” **(Page 9)**

6. p. 10, line 163. What does "normalized" mean here?

We agree this was ambiguous, and we have now clarified:

“... and continuous variables were standardized to a common scale, with a mean of 0 and a standard deviation of 1.” **(Page 10)**

7. Figure 3 and elsewhere. This is a nice way to display these results.

In sum, this paper makes a significant contribution to the literature on variation in longevity. I enjoyed reading it.

We thank the reviewer for their kind words and their helpful suggestions for improvement.